

Voix F2S Literature Survey Document (25 Papers)											
No.	Title	Authors	Year	Theme	Problem Solved	Input → Output	Architecture	Loss / Objective	Key Result	Limitations	Relevance to Project (Voix)
1	Speech2Face: Learning the Face Behind a Voice	Oh, Dekel, Kim, Mosseri, Freeman, Rubinstein, Matusik	2019	Cross-modal voice-to-face inference	Reconstructing a canonical face image from speech alone, without needing to model demographic attributes (age, gender, ethnicity) explicitly.	Complex spectrogram of ~6s speech → 4096-D VGG-Face feature → decoded canonical (frontal, neutral) face image.	Voice encoder (CNN on spectrogram, temporal max-pooling, FC layers) predicting a face feature, fed into a fixed pre-trained face decoder (Cole et al.).	L1 loss on decoder activations + normalized feature loss + knowledge-distillation loss on VGG-Face logits (Eq. 1), $\lambda_1=0.025$, $\lambda_2=200$.	94% gender agreement with true images; R@10 retrieval of 45.82% (cos, 6s audio) vs 10% random, on AVSpeech test set.	Ethnicity performance biased toward "white"/"Asian" due to skewed AVSpeech demographics; language/accent can produce mismatched face predictions; reconstructions resemble "average" faces.	Establishes that voice carries reliable cross-modal signal for face-like attribute inference — motivates VOIX's premise that voice alone can drive visually-grounded outputs, and shows self-supervised (no explicit label) training is viable.
2	Seeing Voices and Hearing Faces: Cross-modal biometric matching	Nagrani, Albanie, Zisserman	2018	Cross-modal biometric matching (face↔voice)	Determining whether static/dynamic face-voice matching is possible "in the wild" beyond controlled human-perception studies.	V-F: audio clip + 2+ face images/videos → select matching face. F-V: face image/video + 2+ voices → select matching voice.	3-stream CNN (2 face sub-nets + 1 voice sub-net, VGG-M) with fusion layer; extended to 5-stream dynamic-fusion (RGB+dynamic image) and N-way "query pooling" architecture.	Cross-entropy loss on binary/N-way position classification of the correct match.	V-F identification accuracy 81.0% (static) / 84.3% (dynamic) on VoxCeleb+VGG Face, exceeding human baseline (81.3% total, but model exceeds humans 63.9% vs 57.1% on GNA-controlled subset).	Some accuracy gains may reflect dataset biases (celebrity recognizability, correlated confounds like profession) rather than pure biometric signal; dynamic gains modest.	Provides direct evidence (beating human performance) that face-voice association is learnable and non-trivial — validates the cross-modal embedding approach VOIX likely builds on for identity consistency between voice and face.
3	ImageBind: One Embedding Space To Bind Them All	Girdhar, El-Nouby, Liu, Singh, Alwala, Joulin, Misra	2023	Multimodal joint embedding learning	Learning a single embedding space across 6 modalities without needing all pairwise combinations of paired data — only image-paired data is required.	Modality-specific inputs (image/video, text, audio, depth, thermal, IMU) → normalized embeddings in one shared space; enables cross-modal retrieval/classification/generation.	Separate Transformer (ViT-based) encoders per modality with modality-specific projection heads, aligned via images as the "binding" modality; image/text initialized from OpenCLIP.	Symmetric InfoNCE contrastive loss between image embeddings and each other's embeddings.	Emergent zero-shot audio classification 17.6% (Audioset), ESC 66.9% Top-1; emergent zero-shot beats/matches specialist models; scaling image encoder (ViT-B→H) improves all modalities.	No paired data between non-image modalities is used, so emergent alignment is only as good as the image encoder; limited to modalities with natural image pairing; specialist supervised models still outperform on downstream tasks.	Demonstrates a scalable recipe for binding arbitrary modalities via one anchor modality — directly relevant if VOIX unifies voice with other modalities (face/video/text) through a shared embedding rather than pairwise training.
4	Imaginary Voice: Face-Styled Diffusion Model for Text-to-Speech	Lee, Chung, Chung	2023	Face-conditioned zero-shot TTS	Enabling multi-speaker TTS for unseen speakers using face images as enrollment instead of clean speech samples, and training face-conditioned end-to-end with the TTS model.	Text transcription + face image → synthesized speech waveform (mel-spectrogram then vocoder).	Grad-TTS-based diffusion acoustic model (text encoder, duration predictor, diffusion noise predictor) conditioned on a face-image speaker embedding from a jointly-trained visual/audio cross-modal biometric network.	Diffusion (score-matching) loss + prior loss + duration loss + speaker feature binding loss (L_{spk} , matching audio-network latent features of synthesized vs. target speech).	5-way cross-modal matching accuracy 38.0% (face-conditioned, unseen speakers) vs 19.4% for Grad-TTS speaker-embedding baseline; MOS ~3.28 for unseen speakers with face condition.	Trained/evaluated on short (~34s/speaker) in-the-wild LRS3 data; still far from upper bound (Mel+HiFi-GAN 48.6% matching); speaker verification/AB tests only ~60% correct under gender-matched hard conditions.	Directly analogous use case to VOIX: using face as a speaker-identity proxy for speech generation, end-to-end trained rather than swapped at inference — a template/baseline for face-conditioned voice synthesis in VOIX.

5	ArcFace: Additive Angular Margin Loss for Deep Face Recognition	Deng, Guo, Xue, Zafeiriou	2019	Face recognition loss function design	Improving discriminative power/geodesic-margin properties of face embeddings over Softmax, SphereFace, and CosFace losses.	Face image $\rightarrow$ normalized 512-D embedding feature (via DCNN, e.g. ResNet50/100).	Standard DCNN backbone (ResNet50/ResNet100) with feature and weight L2 normalization before the loss layer.	Additive Angular Margin Loss: cosine of $(\theta_{yi} + \text{margin})$ scaled by feature scale $s=64$, softmax cross-entropy on top.	99.83% LFW, 98.02% YTF verification with MS1MV2/ResNet100; state-of-the-art on 10 benchmarks incl. MegaFace, IJB-B/C, Trillion-Pairs.	Requires careful hyperparameter tuning (s, m); the margin curve/geometry interpretation adds negligible but nonzero compute; performance ceiling still shown by gaps vs. human fusion baselines on CALFW/CPLFW.	Provides the face-recognition feature extractor commonly used as ground-truth identity supervision (e.g., in Speech2Face, FACE-TTS, DECA) — a foundational component VOIX likely uses/depends on for identity-consistent face embeddings.
6	Learning a model of facial shape and expression from 4D scans (FLAME)	Li, Bolkart, Black, Li, Romero	2017	3D face/head statistical modeling	Building a full-head (not just face-region) statistical model that jointly captures identity shape, pose (incl. jaw/eyeballs/neck), and expression, learned from thousands of scans.	Identity shape β , pose θ , expression ψ parameters $\rightarrow$ 3D head mesh (5023 vertices) via linear blend skinning + blendshapes.	Vertex-based linear model: template mesh + shape blendshapes + pose correctives + expression blendshapes, articulated with linear blend skinning around 4 joints (neck, jaw, eyeballs).	(Model construction/fitting paper — not present in full detail in extracted content; per FLAME's standard training this is a PCA/registration-based fitting objective; not verifiable from the provided excerpt.)	(Only referenced via other papers in the set; full FLAME paper content not present in the provided excerpt to verify numeric results.)	(Not directly verifiable from the excerpt provided.)	FLAME is the underlying 3D geometric prior used by DECA/EMOCA for detailed face reconstruction — relevant to VOIX if 3D animatable head geometry (not just 2D image) is a target output alongside voice.
7	Learning an Animatable Detailed 3D Face Model from In-The-Wild Images (DECA)	Feng, Feng, Black, Bolkart	2021	Animatable detailed 3D face reconstruction	Recovering person-specific, expression-animatable geometric details (wrinkles) from a single in-the-wild image, without paired 3D or high-quality scan supervision.	Single 2D face image $\rightarrow$ FLAME coarse parameters (shape, pose, expression, albedo, camera, lighting) + detail code $\rightarrow$ animatable detailed 3D face mesh with UV displacement map.	ResNet50-based coarse encoder (E_c) + detail encoder (E_d) producing 128-D detail latent; detail decoder (F_d) conditioned on detail code + FLAME expression/jaw-pose to output UV displacement map.	L_coarse = landmark + eye-closure + photometric + identity (via pretrained face recognition net) + shape-consistency + regularization losses; L_detail adds photometric-detail, ID-MRF, soft-symmetry, detail-regularization, and novel detail-consistency loss (L_dc).	State-of-the-art on NoW benchmark: median 1.09mm, mean 1.38mm, std 1.18mm; Feng et al. benchmark 1.48mm median (LQ).	Rendering quality limited by BFM-derived albedo model (bakes in shading); no explicit facial-hair modeling; fails under extreme occlusion/pose; weak perspective camera unsuitable for selfies without focal-length extension.	Introduces the detail-consistency loss and identity-loss recipe (via a pretrained face-recognition net) for disentangling identity from dynamic detail — relevant to VOIX for separating static identity vs. dynamic (voice-driven) expression signals.
8	EMOCA: Emotion Driven Monocular Face Capture and Animation	Daněček, Black, Bolkart	2022	Emotion-faithful 3D face reconstruction	Standard reconstruction losses (landmark, photometric, face-recognition) fail to capture the correct emotional expression from a single image; EMOCA fixes this.	Single 2D image $\rightarrow$ FLAME shape/pose/expression + emotion-consistent 3D face (built on DECA, only expression branch retrained).	DECA architecture with coarse shape encoder fixed (frozen) + new trainable expression encoder (E_e); emotion recognition network (ResNet-50, trained on AffectNet) used only as a fixed perceptual loss network.	$L = \lambda_{\text{emo}} L_{\text{emo}}$ (emotion consistency, L2 distance between emotion-network features of input vs. rendered image) + $\lambda_{\text{pho}} L_{\text{pho}}$ + eye/mouth/lip-corner relative keypoint losses + expression regularizer.	On AffectNet: EMOCA V-CCC 0.77, E-ACC 0.69 — on par with image-based EmoNet (0.73/0.68) and outperforming all other 3D reconstruction baselines (DECA 0.69/0.59). Perceptual study: EMOCA (coarse) 0.68 classification consistency vs DECA 0.33.	Detail (wrinkle) stage doesn't measurably improve perceived emotion and sometimes adds artifacts (lip/forehead); jaw-rotation optimization unstable; inherits DECA's occasional image misalignment.	Demonstrates that a perceptual "consistency loss" against a pretrained recognition network (rather than raw geometric/photometric loss) yields far better semantic fidelity — a methodological pattern VOIX could adopt for training voice-driven facial/expressive outputs to be perceptually (not just geometrically) accurate.

9	ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification	Desplanques, Thienpondt, Demuynck	2020	Speaker embedding extraction for verification	Improving TDNN/x-vector speaker embeddings by incorporating channel attention, multi-scale features, and multi-layer feature aggregation from computer-vision advances.	80-dim MFCC spectrogram (variable-length utterance) → fixed-length (192-D) speaker embedding.	SE-Res2Blocks (1D Res2Net + Squeeze-Excitation) with dilated convolutions, Multi-layer Feature Aggregation (MFA) across blocks, channel- and context-dependent attentive statistics pooling, then FC + AAM-Softmax output.	AAM-Softmax (ArcFace-style additive angular margin) with margin 0.2, scale 30, plus implicit statistics-pooling attention weighting.	EER 0.87% / MinDCF 0.1066 (VoxCeleb1, C=1024 model) — 18.7% relative EER improvement over best baseline (ResNet34/E-TDNN) with fewer parameters (14.7M).	Improvements from context vector to attention are small (~1.9% relative); summed residual connections slightly degrade MinDCF despite improving EER; evaluated only on VoxCeleb-style data.	Provides a strong, efficient speaker-embedding backbone (borrowing ArcFace's AAM-Softmax) that VOIX can use/validate for extracting robust voice-identity representations to pair with face/other modalities.
10	High-Fidelity Audio Compression with Improved RVQGAN (Descript Audio Codec)	Kumar, Seetharaman, Luebs, Kumar, Kumar	2023	Universal neural audio compression/tokenization	Existing codecs (SoundStream, EnCodec) suffer audio artifacts, codebook under-utilization, and are domain-specific; this work builds a universal, high-compression, low-artifact codec.	44.1kHz waveform → discrete tokens (9 codebooks × 10-bit, 86Hz frame rate, 8kbps) → reconstructed waveform via decoder.	Convolutional encoder-decoder with Snake periodic activations, Residual Vector Quantization (factorized + L2-normalized codes), quantizer dropout (p=0.5), multi-period + multi-band multi-scale STFT discriminators.	Multi-scale mel L1 reconstruction loss + HingeGAN adversarial loss + L1 feature-matching loss + VQ codebook/commitment losses (weights 15.0/1.0/2.0/1.0+0.25).	Outperforms EnCodec at all bitrates in MUSHRA listening tests; at 8kbps: ViSQOL 4.18, SI-SDR 10.75 vs EnCodec's 24kbps ViSQOL 3.16, SI-SDR 9.59 — ~90x compression (91.16 factor) of 44.1kHz audio.	Still falls short of reference MUSHRA at highest bitrate; performs worse on environmental sounds than speech; fails to model some instruments (e.g., glockenspiel, synthesizers) well; enables potential deepfake misuse.	If VOIX involves generating or streaming voice output, this codec (or its RVQ/Snake-activation/discriminator design principles) is a candidate high-fidelity, low-bitrate audio representation/tokenization backbone for efficient voice synthesis or transmission.
11	StyleTTS 2: Towards Human-Level TTS through Style Diffusion and Adversarial Training with Large SLMs	Li, Han, Raghavan, Mischler, Mesgarani	2023	Text-to-speech synthesis	Removes StyleTTS's dependence on reference audio and two-stage vocoder training that limited expressiveness/real-time use.	Phonemes (+ optional reference audio) → raw waveform, end-to-end (no separate vocoder).	Text/style encoders, text aligner, pitch extractor, duration/prosody predictors, HiFiGAN/iSTFT Net decoder w/ AdaIN, style diffusion sampler (EDM), MPD/MRD discriminators, WavLM as SLM discriminator.	Lmel (L1) + Ladv/Lfm + Ldur/Lf0/Ln + SLM adversarial loss Lslm via differentiable duration modeling.	CMOS +1.07 vs NaturalSpeech and +0.28 vs ground truth on LJSpeech; near-human on VCTK (CMOS -0.02); MOS 3.83, best among compared models.	Zero-shot similarity on LibriTTS still slightly below Vail-E despite ~250x less training data; diffusion sampling adds inference cost.	Motivates diffusion-sampled style latents + SLM adversarial discriminators for realistic audio without paired reference conditioning at inference.
12	NaturalSpeech 3: Zero-Shot Speech Synthesis with Factorized Codec and Diffusion Models	Ju, Wang, Shen, Tan, Xin, Yang, et al.	2024	Zero-shot TTS via disentangled representation learning	Prior neural-codec TTS entangles content, prosody, timbre, acoustic detail across RVQ levels, hurting quality/similarity/prosody.	Text + prompt speech → factorized codes (content, prosody, timbre, detail) → waveform.	FACodec (Factorized Vector Quantization codec disentangling 4 subspaces) + factorized diffusion model generating duration/content/prosody/detail per prompt.	Information-bottleneck + supervised disentangle-ment losses + adversarial reconstruction (FACodec) + diffusion losses per subspace.	Comparable/better than ground truth CMOS on LibriSpeech; Sim-O 0.64→0.67, SMOS 3.69→4.01; WER 1.94→1.81; scaled to 1B params/200K hrs.	Multi-subspace factorization with separate prompts per attribute increases system/training complexity.	Shows factorizing speech into disentangled latent subspaces (content/prosody/timbre/detail) improves controllability — pattern for VOIX's attribute control.
13	Learning Structured Output Representation using Deep Conditional Generative Models	Sohn, Yan, Lee	2015	Structured output prediction via conditional generative modeling	Deterministic CNNs can't model multi-modal one-to-many mappings (e.g., segmentation); adds stochastic latents for diverse predictions.	Image x (+ optional partial y) → structured label map y sampled via latent z.	Conditional VAE (CVAE): recognition net $q\phi(z x,y)$, conditional prior $p\theta(z x)$, generation net $p\theta(y x,z)$; also GSNN variant + hybrid objective.	$-KL(q\phi(z x,y) \parallel p\theta(z x)) + E[\log p\theta(y x,z)]$; hybrid = $\alpha L_{CVAE} + (1-\alpha)L_{GSNN}$.	GSNN(msc,NI) 93.91% pixel acc/85.39 IoU on CUB test; CVAE(importance sampling) negative CLL 63.91 vs NN's 99.75 on MNIST (1 quadrant).	Training (reconstructs y) differs from testing (prior-only sampling), causing train/test discrepancy — motivated the GSNN fix.	Foundational CVAE formulation underlying conditional generative/diffusion approaches VOIX may use for diverse, stochastic structured predictions.

14	β -VAE: Learning Basic Visual Concepts with a Constrained Variational Framework	Higgins, Matthey, Pal, Burgess, Glorot, Botvinick, Mohamed, Lerchner	2017	Unsupervised disentangled representation learning	No principled unsupervised method learns interpretable factorized latents without prior knowledge of generative factors; more stable than InfoGAN.	Raw image $x \rightarrow$ reconstructed image $\hat{x}$ via disentangled latent z (supports latent traversal).	Modified VAE (encoder $q\phi(z x)$, decoder $p\theta(x z)$) with extra β coefficient weighting the KL term.	$L(\theta, \phi; x, z, \beta) = E[\log p\theta(x z)] - \beta \cdot \text{DKL}(q\phi(z x) \ p(z)), \beta > 1$.	Outperforms VAE, InfoGAN, DC-IGN on disentangleme nt (celebA, 3D chairs, 3D faces) qualitatively and via proposed quantitative metric.	High β trades reconstruction fidelity for disentangleme nt (blurrier output); no closed-form way to pick optimal β .	Provides the β -regularized ELBO mechanism VOIX can tune to disentangle latent voice/style factors (e.g., identity vs. prosody).
15	Flow Matching for Generative Modeling	Lipman, Chen, Ben-Hamu, Nickel, Le	2023	Simulation-free training of continuous normalizing flows	No scalable simulation-free way to train CNFs beyond diffusion-specific score matching; prior simulation-free methods are intractable/biased.	Noise $x_0 \sim N(0, I) \rightarrow$ data sample x_1 via ODE integration of learned vector field vt .	U-Net vector-field regressor $vt(x; \theta)$ trained via Conditional Flow Matching; supports Gaussian paths incl. Optimal Transport (OT).	$LFM(\theta) = E_{t, pt(x)} \ vt(x) - ut(x) \ ^2$, made tractable via per-sample Conditional Flow Matching.	FM-OT best NLL/FID/NFE on CIFAR-10 and ImageNet 32/64/128 vs DDPM, Score Matching, ScoreFlow; faster training/sampling via straight-line paths.	Their CIFAR-10 FID higher than prior works (architecture not optimized for CIFAR-10); validated mainly on image domains.	Offers a faster, more stable alternative to diffusion (straight OT paths) VOIX can adopt for efficient generative sampling of audio/latent reps.
16	High-Resolution Image Synthesis with Latent Diffusion Models	Rombach, Blattmann, Lorenz, Esser, Ommer	2022	Efficient high-resolution image synthesis	Pixel-space diffusion is computationally expensive (hundreds of GPU-days); needs a cheaper, perceptually-equivalent training space.	Image x (+ optional text/semantic-map/bbox conditioning) $\rightarrow$ generated image via decoded latent z .	Pretrained autoencoder (E,D) producing compressed latent $z=E(x)$; U-Net diffusion model in latent space with cross-attention conditioning ($\tau\theta$).	$LLDM = E \ \epsilon - \epsilon\theta(zt, t) \ ^2$; conditional: $E \ \epsilon - \epsilon\theta(zt, t, \tau\theta(y)) \ ^2$.	SOTA FID 5.11 on CelebA-HQ; FID 3.60 on class-conditional ImageNet (w/ classifier-free guidance) at far lower compute than pixel-space DMs.	VQ-regularized latents needed for best sample quality despite slightly worse reconstruction than continuous (KL) variants; SNR of latent space affects conv. sampling beyond trained resolution.	Directly informs VOIX's compressed-latent generative pipeline (encode $\rightarrow$ diffuse $\rightarrow$ decode) and cross-attention conditioning for controllable generation.
17	Morphology and development of the human vocal tract: A study using MRI	Fitch, Giedd	1999	Human vocal tract anatomy and development	Lack of large-sample, non-invasive data testing correlations between vocal tract length (VTL), body size, sex, and puberty.	MRI midsagittal scans (129 subjects, ages 2–25) $\rightarrow$ VTL/segment measurements correlated with height, weight, age, sex, pubertal stage.	N/A — empirical MRI study using "curved line"/"line segment" measurement + regression/ANCOVA/MANOVA statistics.	N/A — least-squares/reduced-major-axis is regression, ANOVA/MANOVA significance testing.	Strong VTL–height correlation ($r=0.926$) and VTL–log body mass ($r=0.941$); sex difference in VTL emerges at puberty (males ~12.9mm longer pharynx).	Sample may overestimate correlation vs. general adult population (large body-size variation); supine scanning may shorten measured VTL vs. upright x-ray.	Empirically grounds VOIX's anatomical assumptions (VTL-body size, sex-based pharynx differences) for physically plausible voice-to-body/face modeling.
18	Correlation Between the Length and Sagittal Projection of the Upper and Lower Jaw and the Fundamental Frequency	Macari, Karam, Tabri, Saredidine, Hamdan	2014	Craniofacial-acoustic correlation (jaw vs. pitch)	Little prior work links facial skeletal (jaw) measurements to F0; tests whether jaw length/position correlates with pitch.	Cephalometric measurements (SNA, SNB, ANB, mandible/maxilla length) + voice recording $\rightarrow$ F0/habitual frequency correlations.	N/A — clinical study; linear regression relating cephalometric variables to F0/habitual frequency (Visi-Pitch IV).	N/A — regression correlation analysis (Pearson r , p -values).	Moderate negative correlation: F0/habitual frequency vs. mandible length ($r=-0.528/-0.577$) and maxilla length ($r=-0.473/-0.519$); angular measures poorly correlated ($r<0.3$).	Small sample (45 subjects, heterogeneous ages); lacks vocal tract/fold length data to substantiate mechanism; social/psychological F0 factors uncontrolled.	Supports plausibility of jaw-length-to-pitch correlations VOIX may exploit when inferring/conditioning voice pitch from facial geometry.
19	Rethinking Voice-Face Correlation: A Geometry View	Li, Wen, Yang, Wang, Singh, Raj	2023	Voice-to-3D-face geometry reconstruction	Prior voice-face work relies on coarse semantics (gender/age/emotion); unclear if 3D geometry is predictable from voice without semantic bias.	Voice recording $\rightarrow$ predicted anthropometric measurements (AMs) $\rightarrow$ reconstructed 3D facial mesh guided by predictable AMs.	Voice-AM-Face pipeline: CNN voice-code encoder + uncertainty-aware AM estimator, optional diffusion-based phonatory module (cross-attention U-Net), AM-guided optimization over low-dimensional face space.	Per-AM uncertainty-weighted regression/KL loss; diffusion loss $\ \epsilon - \epsilon\theta \ _1$; reconstruction loss $\lambda \ \beta \ _2 + \sum (Q_k(P\beta) - \hat{m}(k))^2 - z(k)$.	Hypothesis testing identifies statistically predictable AMs (mostly nose/mouth region); female subset outperforms male; phonatory module cuts error by 0.102 for predictable AMs.	Many AMs (e.g., cranium-wide) not shown predictable with current data/models; non-predictability isn't proven, could reflect modeling/data limits.	Methodological precedent for VOIX's voice-to-face correlation features — shows which facial regions are recoverable from voice, guarding against spurious semantic shortcuts.

20	Learnable PINs: Cross-Modal Embeddings for Person Identity	Nagrani, Albanie, Zisserman	2018	Cross-modal (face-voice) identity embedding	Face/voice recognition treated independently; no self-supervised way to jointly embed them for cross-modal retrieval of unseen identities.	Face image + voice segment (unlabeled talking-face video pairs) → joint 256-D embeddings for face↔voice verification/retrieval.	Two-stream CNN (face subnet f0, voice subnet g0) mapping to shared embedding space, trained with curriculum-based hard-negative mining.	Contrastive loss over positive (same identity/video) and negative (different video) pairs; curriculum controls negative difficulty.	Cross-modal verification AUC 73.8% (unseen-unheard, pretrained init) vs near-chance for random weights; beats SVHF-Net on N-way forced matching for all N.	Same-video pairs score slightly higher (AUC 74.5–74.6 vs 73.8), indicating minor non-identity correlations; gender dominates demographic effect.	Validates the self-supervised contrastive + curriculum training recipe for VOIX's face-voice joint embedding/cross-modal retrieval components.
21	VoxCeleb2: Deep Speaker Recognition	Chung, Nagrani, Zisserman	2018	Large-scale speaker recognition dataset & models	Lack of large-scale, ethnically diverse, freely available audio-visual speaker recognition datasets comparable to face datasets.	Raw audio spectrogram → fixed-length (512-D) speaker embedding for verification/identification.	VGGVox: VGG-M or ResNet-34/50 trunk CNN + pooling → embedding; softmax pretraining then contrastive fine-tuning w/ hard-negative mining.	Cross-entropy softmax pretraining, then contrastive loss (margin α) with offline hard-negative mining.	Dataset: 1M+ utterances, 6,112 speakers, 145 nationalities; ResNet-50 (contrastive) achieves EER 3.95%/Cdet 0.429 on VoxCeleb1 test (vs VGG-M's 7.8% EER).	In the wild data has label/background noise; thresholds set loosely for VoxCeleb2 (training-only), requiring VoxCeleb1 for clean evaluation.	Supplies large-scale diverse training data and ResNet-based speaker embedding architecture VOIX can use for voice-identity representation learning.
22	Looking to Listen at the Cocktail Party: A Speaker-Independent Audio-Visual Model for Speech Separation	Ephrat, Mosseri, Lang, Dekel, Wilson, Hassidim, Freeman, Rubinstein	2018	Audio-visual speech separation	Audio-only separation suffers the label permutation problem and struggles with multi-speaker+noise; prior AV methods were speaker-dependent.	Video (face thumbnails per speaker) + noisy mixed audio (STFT) → per-speaker complex spectrogram mask → isolated waveform.	Dilated-conv visual streams (per face) + dilated-conv audio stream → concatenated features → BLSTM + FC → complex mask per speaker.	L2 squared error between power-law compressed clean spectrogram and enhanced spectrogram (complex ratio mask).	SDR improvement up to 10.6 dB (2 speakers+noise); outperforms SOTA audio-only PIT baseline; near-perfect discrimination of AI-resynthesized vs. original speech.	Low-SNR scenarios ("noisy bar") degrade output quality (masking-based limitation); model isn't real-time, suited to offline post-processing.	Demonstrates a speaker-independent AV fusion architecture (dilated conv+BLSTM+masking) and clean-dataset pipeline (AVSpeech) for VOIX's multi-modal speech tasks.
23	Demographic Fairness in Multimodal Biometrics: A Comparative Analysis on Audio-Visual Speaker Recognition Systems	Fenu, Marras	2022	Fairness/bias in multimodal (face+voice) authentication	Unclear how fusion policies (feature/model/score/decision-level) affect demographic (gender) disparity in AV speaker recognition.	Face+voice biometric example pairs → binary authentication decision (same/different) via one of 7 fusion policies.	Two ResNet-50 unimodal models (face 112×112×3; voice 512×300×1 spectrogram) fused via 7 policies (Feature/Model/Score-Avg-Max-Min/Decision-AND-OR).	N/A new loss — EER-optimal thresholds per policy; formalizes "algorithmic discrimination" as performance disparity across gender classes.	Model-step fusion gives lowest EER on all 3 datasets and lowest gender disparity (e.g., gap 5.8→0.3 on VoxCeleb1-Test vs. voice-only).	Even the best fusion policy shows uneven error patterns by gender depending on which unimodal system was right/wrong; only gender studied (not age/ethnicity).	Provides an empirically validated fusion strategy (model-level fusion) VOIX can adopt to jointly improve accuracy and reduce demographic bias.
24	Svarah: Evaluating English ASR Systems on Indian Accents	Javed, Joshi, Nagarajan, Sundaresan, Nawale, Raman, Bhogale, Kumar, Khapra	2023	ASR benchmark for accent robustness	Existing ASR benchmarks severely underrepresent Indian accents, so "human parity" claims aren't validated on this diverse population.	9.6 hrs transcribed English audio (117 Indian speakers, 65 districts, 19 languages) → ASR transcriptions evaluated via WER across 8 models.	N/A new model — benchmark + evaluation of Whisper (base/medium/large), Wav2Vec2, HuBERT, WavLM, Data2Vec, NVIDIA Conformer, Azure/Google ASR.	N/A — evaluation study reporting WER, no training loss proposed.	Whisper-large: 7.2% WER on Svarah (smallest gap to native WER); self-supervised models (Wav2Vec2, HuBERT, WavLM, Data2Vec) perform far worse (24.5–33.7%).	Even Whisper-large underperforms on everyday use-case utterances (11.2% WER) due to entity names; large models hard to deploy in resource-constrained settings.	Validates need for accent-diverse evaluation before deploying VOIX's ASR/voice components in Indian-accent contexts; identifies best-generalizing model families.
25	Proactive Detection of Voice Cloning with Localized Watermarking	San Roman, Fernandez, Elsahar, D'fosssez, Furon, Tran	2024	Audio watermarking for AI-generated speech detection	Existing watermarking (e.g., WavMark) isn't detection-optimized, isn't localized, and is too slow (brute-force sync) for real-time/large-scale use.	Raw waveform s → watermarked audio $s_w = s + \delta$, plus sample-level detection probability $D(s) \in [0,1]^T$ + optional attribution bits.	Generator/detector pair (EnCodec-based): generator produces additive watermark δ ; detector outputs per-sample detection logits (+optional k-bit attribution layers).	Perceptual losses (f1, multi-scale Mel, adversarial STFT, novel TF-Loudness) + masked sample-level BCE localization loss + optional decoding loss.	AudioSeal: avg AUC 0.97 (vs WavMark's 0.84); IoU 0.99 for 1s segments (vs 0.35); detection 485× faster, generation 14× faster than WavMark.	White-box attacks (detector access) raise detection error ~80% while preserving quality — detector weights must stay confidential; multi-bit attribution has higher false-attribution rate than WavMark.	Provides a proactive, localized watermarking mechanism VOIX can integrate to mark/verify AI-generated voice output, protecting against misuse while staying fast enough for real-time use.